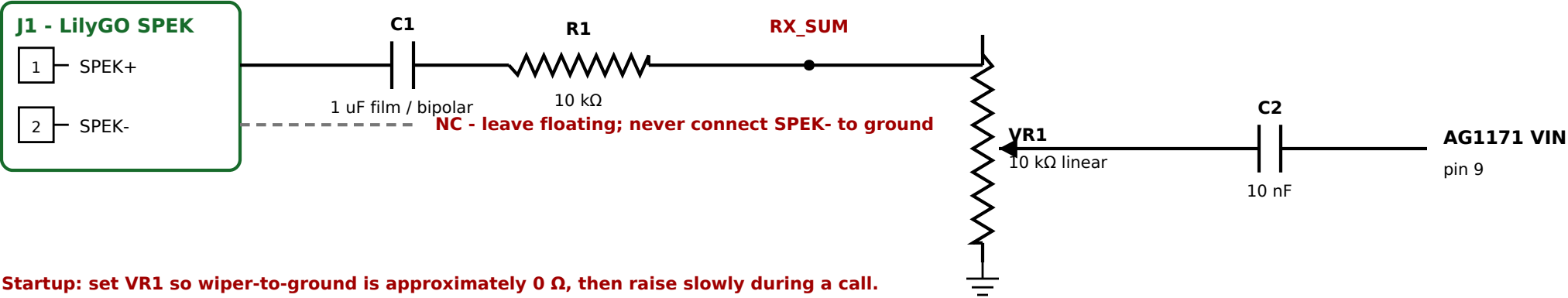
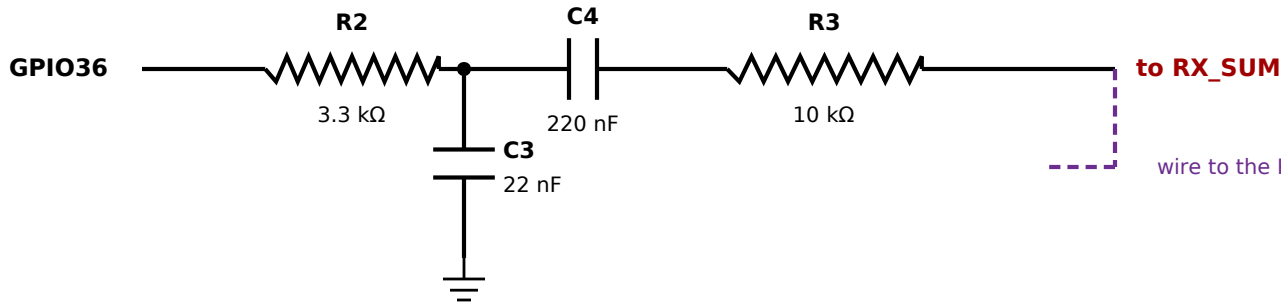


A. RECEIVE: cellular caller -> rotary handset



R1 protects/isolates the BTL output; AG1171 VIN is AC-coupled and level-adjustable.

B. GPIO36 DIAL / BUSY / REORDER TONE INJECTION



Firmware starting settings

PWM carrier: 31 kHz or higher

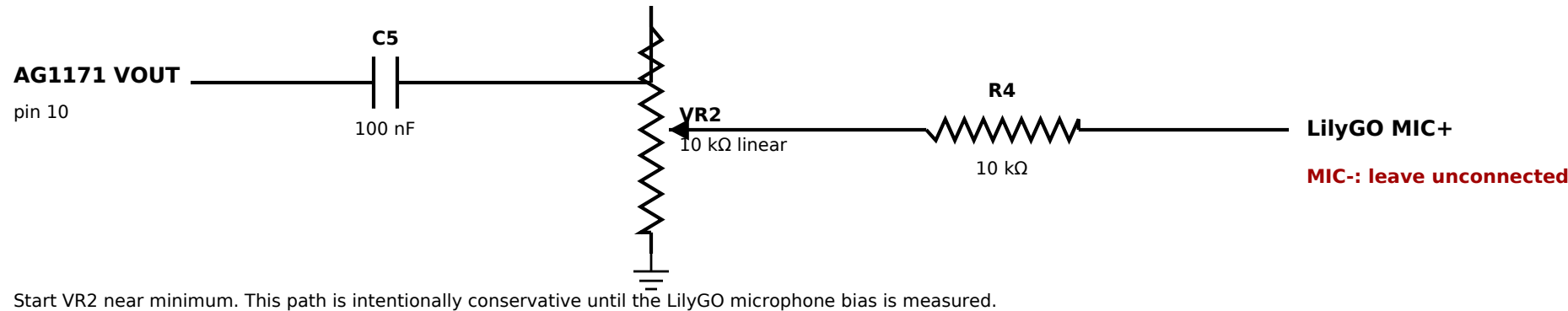
Tone: 350 Hz + 440 Hz

Stop immediately at first dial pulse

R2/C3 are a first-order PWM filter.

C. TRANSMIT: rotary handset -> cellular caller (provisional)

Do not permanently wire MIC- until the three DC measurements below are recorded.



D. REQUIRED DC MEASUREMENTS

Power the LilyGO normally, no call active, audio board disconnected:

1. MIC+ to LilyGO GND
2. MIC- to LilyGO GND
3. MIC+ to MIC-

Record polarity and voltage. These readings determine the final transmit topology.

E. STARTING BOM / BUILD NOTES

VR1, VR2: 10 kΩ linear trimmers

R1, R3, R4: 10 kΩ; R2: 3.3 kΩ

C1: 1 uF film or bipolar

C2: 10 nF; C3: 22 nF; C4: 220 nF

C5: 100 nF

J1: 2-pin SPEK connector; pads for MIC+, GND, GPIO36

Keep the cellular antenna and boost/switching inductors away from audio wiring.